Game Theory

Game: Game is defined as an activity among two or more persons as per set of rules at the end of which each person gets some benefit or bears loss. The set of rules and procedure defines the game. A competitive situation is called game. The term game represents a conflict between two or more parties.

Game Theory: Game theory studies interactive decision-making where the outcome for each participant or players depend on the actions of all.
If you are a player in such a game, when choosing your course of action or strategy, you must take into account the choices of others.

Use of Game Theory : Game theory can help companies make strategies choice within or outside of their organizations, especially against compititors.

Different situations are presented through simple games that set up hypothetical scenario meant to simulate real-world conditions and predict a players behaviour.

Game theory can help to predict how people behave when they are in a competitive situation.

Types of Games :

(1) **Two-person and n-person games :** In two-person games, the players may have many possible choices open to them for each play of the game but the number of players remain only two. Hence, it is called a two person game. In case of more than two person, the game is generally called n-person game.

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(ii) Zero sum game : A zero-sum game is one in which the sum of the payments to all the competitors is zero, for every possible outcome of the game if the sum of the points won, equals the sum of points lost.

(iii) Two person zero sum: A game with two players, where the gain of one player equals the loss of the other, is known as a two-person zero-sum game. It is also called a rectangular game because their pay-off matrix is in rectangular form.

A	B	
+10	-10	= 0

A	B	C	D	E	
+40	-10	-10	-10	-10	= 0

Strategy : The term 'Strategy' is defined as a complete set of plans of action specifying precisely what the player will do under every possible future contigency that might occur during the play of the game, i.e., strategy of a player is the decision rule he uses for making a choice, from his list of courses of action.

Strategy can be classified as :

(i) Pure strategy (ii) Mixed strategy.

Pure strategy : A strategy is called pure if one knows in advance of the play that its is certain to be adopted, irrespective of the strategy the other players might choose.

Mixed strategy : The optimal strategy mixture for each player may be determined by assigning to each strategy, its probability of being chosen. The strategy so determined is called mixed strategy because it is a probabilistic combination of the available choice of strategy. Mixed strategy is denoted by the set S,

$S = \{x_1, x_2, \dots x_n\}$, where x is the probability of choosing course

$$\text{So,} \quad x_1 + x_2 + x_3 + \dots + x_n = 1$$

It is evident that a pure strategy is a special case of mixed strategy.

Pay-off : Pay-off is the outcome of playing the game.
A pay-off matrix is a table showing the amount recieved by the player name at the left hand side after all possible plays of the game. The payment is made by the player named at the top of the table.

If a player A has m courses of action and player B has n courses, then a pay-off matrix may be constructed using the following steps.

(i) Row designation for each matrix are the courses of action available to A.

(ii) Column designation for each matrix are the courses of action available to B.

(iii) With a two-person zero-sum game, the cell entries in B's pay-off matrix will be the negative of the corresponding entries in A's pay-off matrix and the matrices will be as-

$$\begin{array}{cc} & \text{Player B} \\ \text{Player A} & \begin{array}{c|ccccc} & 1 & 2 & 3 & \cdots & n \\ \hline 1 & a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 2 & a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ 3 & a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ m & a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{array} \end{array}$$

$$\begin{array}{cc} & \text{Player B} \\ \text{Player A} & \begin{array}{c|ccccc} & 1 & 2 & 3 & \cdots & n \\ \hline 1 & -a_{11} & -a_{12} & -a_{13} & \cdots & -a_{1n} \\ 2 & -a_{21} & -a_{22} & -a_{23} & \cdots & -a_{2n} \\ 3 & -a_{31} & -a_{32} & -a_{33} & \cdots & -a_{3n} \\ \vdots & \vdots & & & & \\ m & -a_{m1} & -a_{m2} & -a_{m3} & & -a_{mn} \end{array} \end{array}$$

The Maximin - Minimax Principle

This principle is used for the selection of optimal strategies by two players. Considered two players A and B. A is a player who wishes to maximize his gains, while player B wishes to minimize his losses. Since A would like to maximize his minimum gain, we obtain for player A, the value called maximin value and the corresponding strategy is called the maximin strategy.

On the other hand, since player B wishes to minimize his losses, a value called the minimax value, which is the minimum of maximum losses is found. The corresponding strategy is called minimax strategy. when these two are equal (maximin value = minimax value), the corresponding strategies are called optimal strategies and the game is said to have a saddle point. The value of the game is given by the saddle point.

The selection of maximin and minimax strategies by A and B is based upon the so-called maximin - minimax principle, which guarantees the best of the worst results.

<u>Saddle point</u>: A saddle point is a position in the pay-off matrix where, the maximum of row minima coincides with the minimum of column maxima.

The pay-off at the saddle point is called value of the game.

We shall denote the maximin value by $\underline{\gamma}$

the minimax value of the game by $\bar{\gamma}$

and the value of the game by γ

<u>Notes:</u> (i) The game is said to be fair if,

maximin value = minimax value = 0, i.e., $\bar{\gamma} = \underline{\gamma} = 0$

(ii) The game is said to be strictly determinable if,

maximin value = minimax value $\neq 0$. $\underline{\gamma} = \gamma = \bar{\gamma}$.

Q: Solve the game whose pay-off matrix is given below.

Player B

$$\begin{array}{cc|ccccc|c}
 & & B_1 & B_2 & B_3 & B_4 & B_5 & \text{Row minima} \\
 & A_1 & -2 & 0 & 0 & 5 & 3 & -2 \\
\text{Player A} & A_2 & 3 & 2 & 1 & 2 & 2 & 1 \\
 & A_3 & -4 & -3 & 0 & -2 & 6 & -4 \\
 & A_4 & 5 & 3 & -4 & 2 & -6 & -6 \\
\text{Column maxima} & & 5 & 3 & 1 & 5 & 6 &
\end{array}$$

$$\text{Maxi(minimum)} = \underline{\gamma} = \text{Max}(-2, 1, -4, -6) = 1$$

$$\text{Min (Maximum)} = \bar{\gamma} = \text{Min}(5, 3, 1, 5, 6) = 1$$

Since, $\underline{\gamma} = \bar{\gamma} = 1$, there exists a saddle point.

Value of the game $= 1$

The position of saddle point is the optimal strategy and is given by (A_2, B_3)

Q: Determine which of the following two-person zero-sum games are strictly determinable and fair. Given the optimum strategy for each player in the case of strictly determinable games.

(i) Player B

$$\text{Player A}\quad \begin{array}{c|cc|c} & B_1 & B_2 & \\ \hline A_1 & -5 & 2 & -5 \\ A_2 & -7 & -4 & -7 \\ \hline & -5 & 2 & \end{array}$$

$$\text{Min (Maximum)} = \text{Min}(-5, 2) = -5$$

$$\text{Max min} = \text{Max}(-5, -7) = -5$$

$$\bar{\gamma} = \underline{\gamma} = -5 \neq 0$$

∴ The game is strictly determinable.

Value of the game $= -5$

Optimal strategy is the position (A_1, B_1).

(ii) Player B

$$\text{Player A}\quad \begin{array}{c|cc|c} & B_1 & B_2 & \text{Row min.} \\ \hline A_1 & 1 & 1 & 1 \\ A_2 & 4 & -3 & -3 \\ \hline \text{Column max.} & 4 & 1 & \end{array}$$

$$\text{Max (minimum)} = \text{Max}(1, -3) = 1$$

$$\text{Min (Maximum)} = \text{Min}(4, 1) = 1$$

$$\bar{\gamma} = \underline{\gamma} = 1 \neq 0$$

∴ The game is strictly determinable

value of the game $= 1$

Optimal strategy is the position is (A_1, B_2).

Games Without Sadle points (Mixed Strategy)

A game without saddle point can be solved by various solution methods.

(i) 2 × 2 Games without Saddle point

(ii) Graphical Method for 2×n or m×2 games

(iii) Dominance Property.

2×2 Games Without Saddle point

Consider a 2×2 two-person zero-sum game without any saddle point, having the pay-off matrix for player A as,

$$\begin{array}{c} \\ A_1 \\ A_2 \end{array} \begin{array}{c} \begin{array}{cc} B_1 & B_2 \end{array} \\ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \end{array}$$

The optimum mixed strategies, $S_A = \begin{bmatrix} A_1 & A_2 \\ p_1 & p_2 \end{bmatrix}$ and $S_B = \begin{bmatrix} B_1 & B_1 \\ q_1 & q_2 \end{bmatrix}$

where $p_1 = \dfrac{a_{22} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$, $p_1 + p_2 = 1 \Rightarrow p_2 = 1 - p_1$

$q_1 = \dfrac{a_{22} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$, $q_1 + q_2 = 1 \Rightarrow q_2 = 1 - q_1$

The value of the game $(v) = \dfrac{a_{11}a_{22} - a_{12}a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$

Q: Solve the following pay-off matrix. Also determine the optimal strategies and value of the game.

$$\begin{array}{cc} & B \\ A & \begin{bmatrix} 5 & 1 \\ 3 & 4 \end{bmatrix} \end{array}$$

Solution:

let,

$$\begin{array}{ccc} & B_1 & B_2 \\ A_1 & a_{11} & a_{12} \\ A_2 & a_{21} & a_{22} \end{array}, \quad S_A = \begin{pmatrix} A_1 & A_2 \\ p_1 & p_2 \end{pmatrix}, \quad S_B = \begin{bmatrix} B_1 & B_2 \\ q_1 & q_2 \end{bmatrix}$$

$$\therefore p_1 = \frac{a_{22} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{4 - 3}{(5+4) - (1+3)} = \frac{1}{5}$$

$$p_1 + p_2 = 1 \Rightarrow p_2 = 1 - p_1 = 1 - \frac{1}{5} = \frac{4}{5}$$

$$q_1 = \frac{a_{22} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{4 - 1}{(5+4) - (1+3)} = \frac{3}{5}$$

$$q_1 + q_2 = 1 \Rightarrow q_2 = 1 - q_1 = 1 - \frac{3}{5} = \frac{2}{5}$$

$$\text{Value of the game} = \frac{a_{11}\,a_{22} - a_{12}\,a_{21}}{(a_{11}+a_{22}) - (a_{12}+a_{21})}$$

$$= \frac{5 \times 4 - 1 \times 3}{(5+4) - (1+3)} = \frac{17}{5}$$

$\therefore$ Optimum mixed strategy,

$$S_A = \left(\frac{1}{5}, \frac{4}{5}\right) \quad \text{and} \quad S_B = \left(\frac{3}{5}, \frac{2}{5}\right)$$

Q: In a game of matching coins with two players, suppose A wins one unit of value when there are two heads, wins nothing when there are two tails and losses $\frac{1}{2}$ unit of value when there are one head and one tail. Determine the pay-off matrix, the best strategies for each player and the value of the game to A.

Solution: The pay-off matrix for this is given by,

$$\begin{array}{cc|cc|c|c}
 & & \multicolumn{2}{c|}{\text{Player B}} & & \\
 & & H & T & \text{oddments} & \text{prob. (p)} \\
\text{Player A} & H & 1 & -\frac{1}{2} & \frac{1}{2} & \dfrac{\frac{1}{2}}{\frac{1}{2}+\frac{3}{2}} = \frac{1}{2}\times\frac{2}{4} = \frac{1}{4} \\
 & T & -\frac{1}{2} & 0 & \frac{3}{2} & \frac{3}{2}\times\frac{2}{4} = \frac{3}{4} \\
 & \text{oddments} & \frac{1}{2} & \frac{3}{2} & & \\
 & \text{prob. (q)} & \dfrac{\frac{1}{2}}{\frac{4}{2}} = \frac{1}{4} & \dfrac{\frac{3}{2}}{\frac{4}{2}} = \frac{3}{4} & &
\end{array}$$

Optimum strategy, $S_A = \left(\frac{1}{4}, \frac{3}{4}\right)$ & $S_B = \left(\frac{1}{4}, \frac{3}{4}\right)$

$$\text{Value of game} = \frac{\left(1\times\frac{1}{2}\right) + \left(-\frac{1}{2}\times\frac{3}{2}\right)}{\frac{1}{2}+\frac{3}{2}}$$

$$= \frac{\frac{1}{2}-\frac{3}{4}}{4/2} = -\frac{1}{2}\times\frac{1}{4} = -\frac{1}{8}$$

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Graphical Method for $2 \times n$ or $m \times 2$ Games

Q: Solve the following 2×3 game graphically.

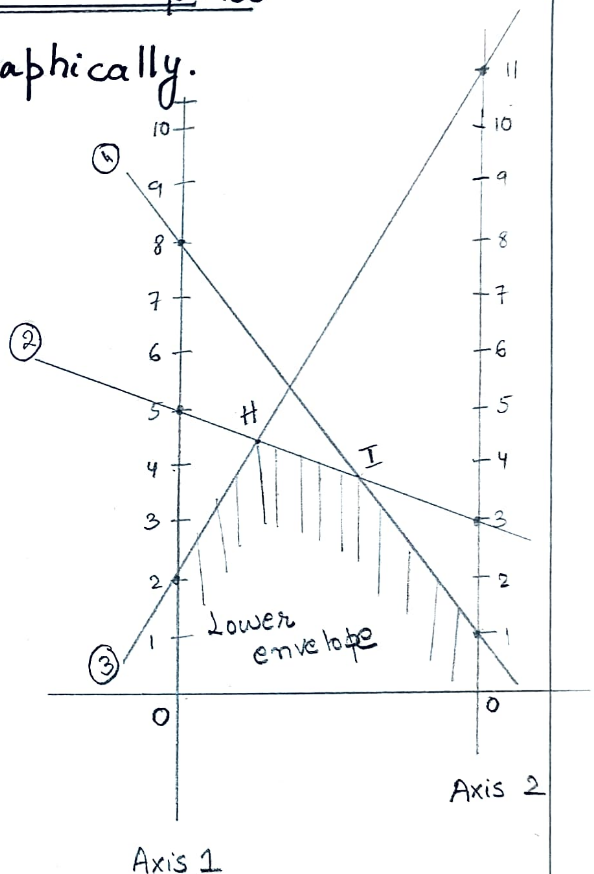
$$\begin{array}{cc} & \text{Player B} \\ \text{Player A} & \left[\begin{array}{ccc} 1 & 3 & 11 \\ 8 & 5 & 2 \end{array}\right] \begin{array}{c} 1 \\ 2 \end{array} \\ & \begin{array}{ccc} 8 & 5 & 11 \end{array} \end{array}$$

Solution: Since the problem does not posses any saddle point, let the player A play by the mixed strategy

$$S_A = \begin{pmatrix} A_1 & A_2 \\ p_1 & p_2 \end{pmatrix} \text{ with } p_2 = 1 - p_1$$

against player B.

The graph plotting for the above pay-off matrix is given by-



Now since player A wishes to maximize his minimum excepted pay-off, we considered the heighest point of intersection H on the Lower envelope of A's expected pay-off equation. The lines II and III passing through H define the relevant moves that II and III alone need to play. The solution to the original 2×3 game reduces to,

$$
\begin{array}{c|cc|c|c}
 & B_1 & B_2 & \text{oddments} & \text{prob.}(p) \\
A_1 & 3 & 11 & -3 & \frac{-3}{-11} = \frac{3}{11} = p_1 \\
A_2 & 5 & 2 & -8 & \frac{-8}{-11} = \frac{8}{11} = p_2 \\
\hline
\text{oddments} & -9 & -2 & & \\
\text{prob}(q) & \frac{-9}{-11} & \frac{-2}{-11} & & \\
 & \frac{9}{11} = q_1 & \frac{2}{11} = q_2 & &
\end{array}
$$

The optimum strategy for A and B is given by,

$$S_A = \left(\frac{3}{11}, \frac{8}{11}\right) \text{ and } S_B = \left(\frac{9}{11}, \frac{2}{11}\right)$$

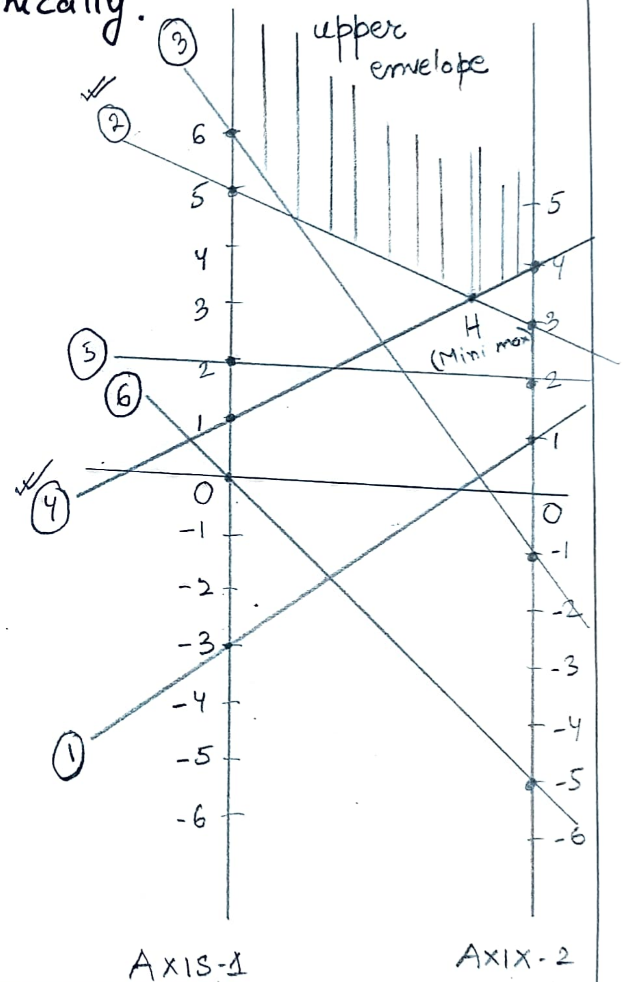
So, value of the game $= \dfrac{(3\times-3)+(5\times-8)}{-3-8} = \dfrac{-9-40}{-11} = \dfrac{-49}{-11}$

$$= \frac{49}{11}$$

Graphical Method for $m \times 2$ Games

Q: Solve the 6×2 game problem graphically.

$$
A \quad \begin{array}{c|cc|c}
 & B_1 & B_2 & \\
A_1 & 1 & -3 & -3 \\
A_2 & 3 & 5 & 3 \\
A_3 & -1 & 6 & -1 \\
A_4 & 4 & 1 & 1 \\
A_5 & 2 & 2 & 2 \\
A_6 & -5 & 0 & -5 \\
 & 4 & 6 &
\end{array}
$$



Solution: The given problem does not posses any saddle point. Therefore, let player B play by the mixed strategy,

$$S_B = \begin{pmatrix} B_1 & B_2 \\ q_1 & q_2 \end{pmatrix} \text{ with } q_2 = 1 - q_1$$

against player A.

Since player B wishes to minimize his maximum expected pay-off, we considered the lowest point of the upper boundary of B's expected pay-off equation. The point H (intersection of lines 2 and 4) represents the minimax expected value of the game for player B. Hence the solution to original 6×2 game reduces to 2×2 pay-off matrix.

$$\begin{array}{c|cc|c|c} & B_1 & B_2 & \text{oddments} & \text{prob.} \\ A_2 & 3 & 5 & -3 & \frac{-3}{-5} = \frac{3}{5} = p_1 \\ A_4 & 4 & 1 & -2 & \frac{-2}{-5} = \frac{2}{5} = p_2 \\ \text{oddments} & -4 & -1 & & \\ \text{prob.} & \frac{-4}{-5} = \frac{4}{5}(q_1) & \frac{-1}{-5} = \frac{1}{5}(q_2) & & \end{array}$$

The optimal strategy for A and B is given by, $S_A = (0, \frac{3}{5}, 0, \frac{2}{5}, 0, 0)$

$S_B = (\frac{4}{5}, \frac{1}{5})$

$$\text{Value of the game} = \frac{(3\times -3) + (4\times -2)}{-3-2} = \frac{-9-8}{-5} = \frac{-17}{-5} = \frac{17}{5}$$

Dominance Property example - 2

Q: Using the principle of dominance, solve the following game.

$$
\text{Player A} \quad
\begin{array}{c}
\text{Player B} \\
\begin{bmatrix} 3 & -2 & 4 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{bmatrix}
\begin{array}{c} 5 \\ 5 \\ 10 \end{array} \\
\begin{array}{ccc} 4 & 4 & 12 \end{array}
\end{array}
$$

Solution: Since all the elements in the third column are greater than or equal to the corresponding elements in the first column. Therefore, column three is dominated by column - 1. So, we can delete column - 3. So the reduced pay-matrix will be given by,

$$
\underset{r \quad c}{3 \times 2} \quad
\begin{array}{c|cc}
 & B_1 & B_2 \\
\hline
A_1 & 3 & -2 \\
A_2 & -1 & 4 \\
A_3 & 2 & 2
\end{array}
$$

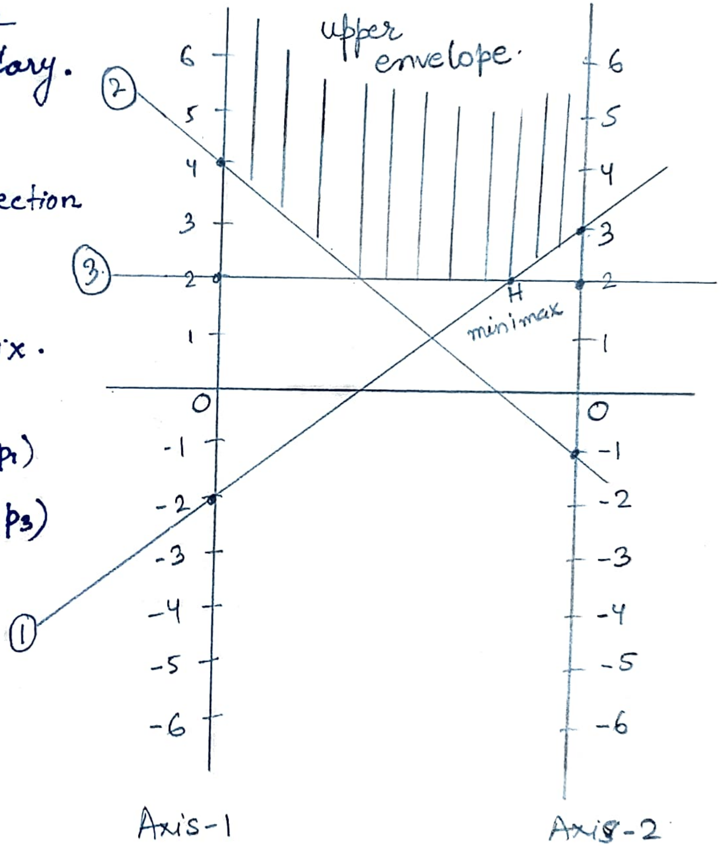
Since no rows and no column is dominates others. The 3×2 game can be solved using graphical method.

Since player B wishes to minimize his maximum loss, we find the lowest point of the upper boundary.

The lowest point in the upper boundary is given by the intersection of lines 1 and 3.

The solution in the original game is reduced to 2x2 matrix.

$$\begin{array}{c|cc|c|c} & B_1 & B_2 & \text{oddments} & \text{prob.} \\ \hline A_1 & 3 & -2 & 0 & \frac{0}{5} = 0 \ (p_1) \\ A_3 & 2 & 2 & 5 & \frac{5}{5} = 1 \ (p_3) \\ \hline \text{oddments} & 4 & 1 & & \\ \text{prob.} & \frac{4}{5} \ (q_1) & \frac{1}{5} \ (q_2) & & \end{array}$$



The optimum strategy for A and B is given by,

$$S_A = \begin{array}{ccc} A_1 & A_2 & A_3 \end{array} \\ \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$$

$$S_B = \begin{array}{ccc} B_1 & B_2 & B_3 \end{array} \\ \begin{pmatrix} \frac{4}{5} & \frac{1}{5} & 0 \end{pmatrix}$$

$$\text{Value of the game} = \frac{(3\times0) + (2\times5)}{0+5}$$

$$= \frac{0+10}{5} = 2$$

Dominance Property

The principle of dominance states that if one strategy of a player dominates over the other strategy in all conditions then the later strategy can be ignored. A strategy dominates over the other only if it is preferable over other in all conditions.

In such cases of dominance, we reduce the size of the pay-off matrix by deleting those strategies, which are dominated by others. The general rule for dominance are:

(i) If all the elements of a row, say K^{th} row, are less than or equal to the corresponding elements of any other row, say r^{th} row, then K^{th} row is dominated by the r^{th} row.

(ii) If all the elements of a column, say k^{th} column, are greater than or equal to the corresponding element of any other column, say r^{th} column, then K^{th} column is dominated by r^{th} column.

(iii) Dominated rows and columns may be deleted to reduce the size of the pay-off matrix as the optimal strategies will remain unaffected.

(iv) If some linear combinations of some rows dominate i^{th} row, then the i^{th} row will be deleted.
Similar arguments follow for columns.

Q: Solve the following game.

$$
\begin{array}{c c c c c}
 & & & \text{Player } B & \\
 & & B_1 & B_2 & B_3 \\
\text{Player } A & \begin{array}{c} A_1 \\ A_2 \\ A_3 \end{array} & \multicolumn{3}{c}{\left[\begin{array}{ccc} 1 & 7 & 2 \\ 6 & 2 & 7 \\ \cancel{5} & \cancel{1} & \cancel{6} \end{array}\right]} & \begin{array}{c} 10 \\ 15 \\ \cancel{12} \end{array}
\end{array}
$$

Solution: Since all the element in the third row are less than or equal to the corresponding elements of second row, therefore the third row is dominated by 2nd row. So we can delete third row (dominated row.)

Now, the reduced pay-off matrix is

$$
\begin{array}{c c c c}
 & B_1 & B_2 & B_3 \\
\begin{array}{c} A_1 \\ A_2 \end{array} & \multicolumn{3}{c}{\left[\begin{array}{ccc} 1 & 7 & \cancel{2} \\ 6 & 2 & \cancel{7} \end{array}\right]} \\
 & 7 & 9 & \cancel{9}
\end{array}
$$

The elements of third column are greater than or equal to the corresponding elements of the first column, which gave that column 3 is dominated by column 1. So we can delete column-3.

$$\begin{array}{c|cc|c} & B_1 & B_2 & \text{oddments} \\ A_1 & 1 & 7 & -4 \\ A_2 & 6 & 2 & -6 \\ \hline \text{oddments} & -5 & -5 & \end{array}$$

prob. (p)

$$\frac{-4}{-4-6} = \frac{-4}{-10} = \frac{2}{5} \ (p_1)$$

$$\frac{-6}{-10} = \frac{3}{5} \ (p_2)$$

prob (q) $\quad \frac{-5}{-10} \quad \frac{-5}{-10}$

$$\frac{1}{2} (q_1) \quad \frac{1}{2} (q_2)$$

Optimal strategy is given by,

$$S_A = \begin{bmatrix} A_1 & A_2 & A_3 \\ \frac{2}{5} & \frac{3}{5} & 0 \end{bmatrix}$$

$$S_B = \begin{bmatrix} B_1 & B_2 & B_3 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

Value of the game $= \dfrac{(1 \times -4) + (6 \times -6)}{-4-6} = \dfrac{-4-36}{-10} = \dfrac{-40}{-10} = 4$